Pengcen Jiang

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Education

Ph.D. Data Science, University of California, San Diego (UCSD) Sep. 2022 – Present, expected June 2027
M.S. Data Science (en route), University of California, San Diego (UCSD) June 2026
B.S. Physics, University of Science and Technology of China (USTC) Sep. 2017 – June 2022

- School of the Gifted Young, 5-year Interdisciplinary Talent Program, **double major in Stats & Applied Maths**.

Technical Skills

Programming Languages: 7+ years of experience in Python and MATLAB; proficient in R, C.
Engineering Skills: PyTorch/PyTorch Lightning, TensorFlow, NumPy, Pandas, SciPy, Git, Linux, Kubernetes, Hugging Face, scikit-learn, Matplotlib, Seaborn, LaTeX, Scanpy; Large-scale experiment automation, parallel/multi-threaded/multi-process computing and programming.
Machine Learning & Data Science: Deep learning (RNN/LSTM/GRU, CNN, attention, Bayesian NN), continual learning, generative models, supervised/unsupervised learning, model interpretability, dimensionality reduction, manifold learning, control theory, (non)linear dynamical systems, time series analysis, hypothesis testing, anomaly detection.

Industry Experience

Analytic Science Intern, FICO (Manager: Joseph Murray) June 2026 – Sep. 2026
Topic: Unsupervised learning and synthetic data generation for network intrusion detection

- Co-developed the notebooks and datasets for the 4th FICO Educational Analytics Challenge, reaching nine universities and >300 undergraduate students.
- Raised cyberattack detection on network traffic data (CIC-IDS-2017) from ∼**10% to** ∼**85%** at a 5% false alarm rate (overall AUC from 0.70 to 0.865) by redesigning the reference distribution of the unsupervised anomaly detector, Classifier-Adjusted Density Estimation (CADE), as a mixture of Gaussian KDE and uniform distribution in PCA space.
- Built synthetic network traffic generators (TabSyn, flow matching) and brought synthetic data **98% closer** to real traffic (energy distance reducing from 0.48 to 0.011) by conditioning on traffic type.

Research Experiences

Graduate Student Researcher (Advisor: Dr Mikio Aoi, UCSD) Sep. 2022 – Present
Topic: Interpretability of Artificial and Biological Neural Networks

- Trained student neural networks with **multi-modal** data of **multi-task** teachers (synthetic task-trained RNN) that aligned sensory, neural, and behavioral signals with **continual learning** and **regularization**.
- **Led a four-member research team**, coordinating analysis and ensuring fast, high-quality results.
- Managed **large-scale experimentation** (>1000 models) with **automated** sweeping, training, and evaluation.
- Cut runtime of two open-source analysis packages (Dynamical Similarity Analysis, FixedPointFinder) from **days to under 1 hour** by parallelizing computation.
- Showed that matching behavior is not enough to recover a model's internal mechanism, and found a control-theory measure (gramians) that predicts which models do recover it.
- Showed with Schur decomposition that the dominant activity patterns of the fruit fly olfactory circuit are consistent across five brain connectomes.

Undergraduate Research Assistant (Advisor: Dr Quan Wen, USTC) Jan. 2022 – June 2022
Topic: Statistical inference of connectome from zebrafish whole-brain calcium imaging

- Fit **generalized linear models (GLMs)** to whole-brain calcium imaging of larval zebrafish to infer neural connectivity; predicted connectivity patterns matched both experimental data and an RNN benchmark.

Undergraduate Summer Research Intern (Advisor: Dr Guangyu Robert Yang, MIT) July 2021 – Dec. 2021
Topic: Design and analysis of flexible working memory models processing naturalistic inputs

- Designed **multi-region** neural networks (CNN, RNN, attention) and cognitive task simulations; trained models matched animal behavior and neural dynamics.

Selected Publications

- (in prep) **P. Jiang**, H. Giaffar, Z. Jin, A. Charles, M. Aoi, Understanding degeneracy in multi-task RNNs using controllability and observability Gramians along a trajectory. (2026)
- (in prep) H. Giaffar, **P. Jiang**, et al., Neural connectivity constrains computation in the insect Lateral Horn. (2026)
- Yudi Xie, Yu Duan, Aohua Cheng, **Pengcen Jiang**, Christopher J. Cueva, and Guangyu Robert Yang. Natural constraints explain working memory capacity limitations in sensory-cognitive models. bioRxiv (2023)

- **Pengcen Jiang**, Yu Hu, Quan Wen. Inferring Connectome Based on Whole-Brain Calcium Fluorescence Imaging Data of Zebrafish. University of Science and Technology of China Undergraduate Thesis (2022)
- Fengchao Wang, **Pengcen Jiang**, Qiong Fan. Capillary condensation under nano-confinement: A supplement to the 150-year-old Kelvin equation. Chinese Science Bulletin (2021)

Conference Presentations

- **P. Jiang**, et al., Indoctrination is harder than you think: On the use and misuse of RNNs as surrogates for the brain. 2026 Conference on Computational and Systems Neuroscience (COSYNE)
- **P. Jiang**, et al., Indoctrinating students is harder than you think: highly constraining RNNs is no guarantee of their quality as surrogates for the brain. 2025 Conference on Society for Neuroscience (SfN)
- **P. Jiang**, et al., Indoctrinating students is harder than you think: highly constraining RNNs is no guarantee of their quality as surrogates for the brain. 2025 Statistical Analysis of Neural Data Workshop (SAND)
- H. Giaffar, **P. Jiang**, et al., How does connectivity constrain computation in the insect Lateral Horn? 2025 Statistical Analysis of Neural Data Workshop (SAND)
- **Pengcen Jiang**, Adam Charles, Mikio Aoi. Recurrent Neural Networks Utilize Different Mechanisms when Learning Kalman Systems. 2024 Conference on Society for Neuroscience (SfN)
- Y. Xie, ..., **P. Jiang**, C. Cueva, G. R. Yang. Human-like capacity limits in working memory models result from naturalistic sensory constraints. 2023 Conference on Computational and Systems Neuroscience (COSYNE).
- Y., Xie, Y. Duan, A. Cheng, **P. Jiang**, C. Cueva, G. R. Yang. Human-like capacity limitation in multi-system models of working memory. 2022 Conference on Cognitive Computational Neuroscience (CCN).

Selected Honors and Awards

Graduate & Professional Student Association (GPSA) Fall Travel Grant Award, UCSD	2025
Yearly English News Reporter, USTC	2020
Grand Prize (top 14 teams out of 100+ teams) in the 15th Competition of Physical Research Experiments, USTC	2019
Excellent Student Scholarship-Silver, USTC	2019
Cyrus Tang Foundation Moral Education Scholarship, USTC	2017 – 2021
Four-Star Volunteer of Chinese Youth Volunteer Service (300 hrs+ volunteer service), USTC	2019
Outstanding Freshman Scholarship, USTC	2017

Teaching Experiences

DSC212 Probability and Statistics for Data Science, UCSD	SP 2025, FA 2026
DSC140A Probabilistic Modeling and Machine Learning, UCSD	SP 2026
DSC20 Programing/DataStructure for Data Science, UCSD	WI 2026
DSC257R Unsupervised Learning, UCSD	FA 2025
Introduction to Computational Neuroscience, USTC	FA 2021

Academic Activities

Reviewer, Conference on Computational and Systems Neuroscience (COSYNE)	2025-2026
Invited speaker, Research Experiences for Undergraduates, California State University, Northridge	July 2025
Volunteer, Brain Machine Interface Workshop for Undergrads, UCSD	Sep. 2023
Student, CNeuro Summer School, Tsinghua University	2020

Outreach & Leadership

Mentor at Graduate Women in Computing (GradWIC), UCSD	2023 – 2026
Leader of Feiyue, a student-run platform providing information for students to study abroad, USTC	2022
Assistant editor and scientific research journalist at English News Center, USTC	2019 – 2022
Leader of a student scientific expedition group to Maowusu Desert, investigating desert control, USTC	2019
Leader of Special Education Service Group at Fang Cao Young Volunteers Association, USTC	2018 – 2019